

UNIT I: Basic Concepts, OSI/TCP-IP Models, Interfaces and Modems

1.1 Components of Data Communication

Definition: Data communication is the exchange of data between two devices through a transmission medium (wire or air).

Easy Meaning: Sending information from one device to another using a cable or a wireless path.

Component	Meaning	Example
Message	The information to be sent (text, number, image, audio, video)	E-mail, photo
Sender	Device that sends the message	Computer, mobile, camera
Receiver	Device that receives the message	Computer, printer, TV
Transmission medium	Physical path on which the message travels	Twisted pair, fibre, radio waves
Protocol	Set of rules that both devices follow. Without it, devices are connected but cannot understand each other (like two people speaking different languages)	TCP/IP, HTTP

Key Points (qualities of good data communication):

- **Delivery** - data reaches the correct destination only.
- **Accuracy** - data is delivered without change (no errors).
- **Timeliness** - data is delivered in time (very important for live audio/video).
- **Jitter** - uneven delay in arrival of packets; it should be low.

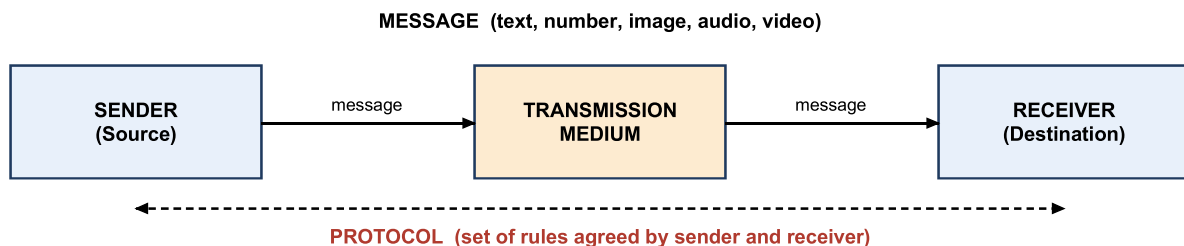


Fig 1.1: Five components of data communication

How to Write in Exam: 1) Definition 2) List five components with a table 3) Diagram 4) Four qualities (delivery, accuracy, timeliness, jitter).

Keywords: message, sender, receiver, medium, protocol, delivery, accuracy, timeliness, jitter

1.2 Distributed Processing

Definition: Distributed processing is a method in which a task is divided among many computers instead of one central machine doing everything.

Simple Explanation: Each computer does a part of the work. Most networks today use it: a big database or job is shared among many machines.

Example: World Wide Web: web servers, database servers and your browser share the work of showing one web page.

- **Security/encapsulation** - a user can access only the part of a system he is allowed to.
- **Distributed databases** - no single machine holds all the data.
- **Faster problem solving** - many computers work on a problem together.
- **Collaborative processing** - many computers work together (e.g. multiplayer games).

Advantages: Better speed, less load on one machine, reliability (if one fails, others work), easy expansion.

Disadvantage: Complex to manage and needs a good network.

1.3 Standards and Organizations

Definition: A standard is an agreed set of rules/specifications so that devices of different makers can work together (interoperability).

Type of standard	Meaning	Example
De facto ("by fact")	Not officially approved, but widely used in practice	TCP/IP (started in practice), Windows
De jure ("by law")	Officially approved by a recognized body	OSI model (ISO)

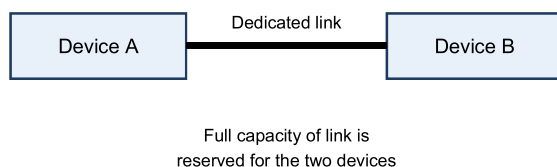
Organization	Full form	Role
ISO	International Organization for Standardization	Made the OSI model; global standards
ITU-T	International Telecommunication Union - Telecommunication Standardization Sector (earlier CCITT)	Standards for telephone and data communication (e.g. X.21, V.90)
ANSI	American National Standards Institute	Official US standards body
IEEE	Institute of Electrical and Electronics Engineers	LAN standards (802 series: Ethernet 802.3, Wi-Fi 802.11)
EIA	Electronic Industries Alliance	Interface standards such as RS-232 (EIA-232)
IETF	Internet Engineering Task Force	Internet protocols (documents called RFCs)
FCC	Federal Communications Commission (USA)	Regulatory agency for radio, TV, telephone

1.4 Line Configuration

Definition: Line configuration is the way two or more devices are connected to a link. A link is the communication path between devices.

Point	Point-to-Point	Multipoint (Multidrop)
Meaning	Dedicated link between exactly two devices	One link shared by three or more devices
Capacity	Full capacity used by the two devices	Shared: spatially (used together) or time-shared (turn by turn)
Cost	More cables, costlier	Less cable, cheaper
Example	Computer to printer; remote control to TV	Mainframe (host) with many terminals

(a) Point-to-Point



(b) Multipoint (Multidrop)

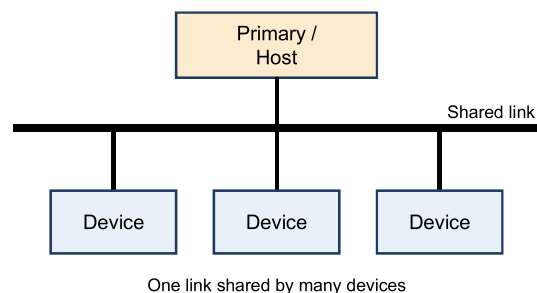


Fig 1.2: Line configurations

1.5 Topology

Definition: Topology is the geometric arrangement of links and nodes (devices) in a network.

Simple Explanation: It shows how computers are connected to each other: as a line, star, circle, etc.